

## **Overview**

The 4×4 Sudoku game is a simple logic-based puzzle game designed to enhance problem solving and critical thinking skills. The game consists of a 4×4 grid where players must fill the empty cells with numbers from 1 to 4. The objective is to ensure that each row, each column and each 2×2 sub grid contains all numbers from 1 to 4 without repetition. The Sudoku Game is a simple number puzzle game. Players fill the empty boxes with numbers following the game rules. The goal is to complete the puzzle correctly. This project helps improve logical thinking, concentration and problem solving skills. It is easy to play and enjoyable for everyone. The Sudoku Game is a simple number puzzle game. Players fill the empty boxes with numbers following the game rules. The goal is to complete the puzzle correctly. This project helps improve logical thinking, concentration and problem solving skills. It is easy to play and enjoyable for everyone.

## **Objectives**

The main objective of this project is to develop a simple 4×4 Sudoku game that helps users improve their logical thinking and problem solving skills. The project aims to create a user friendly game where players can fill empty cells with correct numbers according to Sudoku rules and check their solutions.

Objectives are :

- 1 .To design and implement a basic Sudoku puzzle game.
- 2 . To understand game logic and problem-solving techniques.
3. To create an interactive user interface for players.
4. To validate user inputs and check correct answers.
5. To improve programming and algorithm development skills.

## **Introduction**

Sudoku is a well known logic based number puzzle game that requires players to use their thinking ability and problem solving skills to complete a puzzle. In this project a simple 4×4 Sudoku Game is designed and developed to provide an easy and enjoyable gaming experience for users. The player needs to fill the empty cells with numbers from 1 to 4 while following the basic Sudoku rules where no number can be repeated in the same row, column or grid.

The main goal of this project is to develop a small interactive game that demonstrates the use of programming logic, algorithms and user input handling. The system checks the players answers and determines whether the completed puzzle is correct or incorrect. This project helps to understand the fundamentals of game development and improves programming skills.

The Sudoku Game is designed with a simple interface so that users can easily understand and play the game. It also helps users improve their logical reasoning, concentration and decision making abilities. This project represents the practical application of programming knowledge in creating an engaging and useful application.

### **Scope of sudoku game project**

The scope of this project is to develop a simple and interactive 4×4 Sudoku Game that allows users to solve number puzzles using logical thinking and problem solving skills. The project focuses on implementing the basic rules of Sudoku creating an easy to use interface and checking the correctness of user inputs.

The system will provide a puzzle with some pre filled numbers and allow players to complete the empty cells with the correct numbers. It will validate the entered values and display the result based on whether the solution is correct or not.

This project covers:

- 1 .Designing a basic Sudoku game interface.
- 2 .Generating and displaying 4×4 Sudoku puzzles.
3. Accepting user input for empty cells.
4. Checking Sudoku rules and validating solutions.
5. Providing feedback after puzzle completion.
6. Improving programming, logical thinking and algorithm development skills.

### **Basic features of sudoku game project**

1. User friendly Interface: Provides a simple and easy to use interface for players.
2. 4×4 sudoku puzzle: Allows users to solve a basic 4×4 Sudoku puzzle.
- 3 .Number input system: Users can enter numbers into empty cells.
- 4 .Rule validation: Checks whether the entered numbers follow Sudoku rules.
- 5 .Solution checking: Determines whether the completed puzzle is correct or incorrect.
- 6 .Error detection: Identifies wrong inputs and helps users correct mistakes.
7. Puzzle display: Shows pre filled numbers and empty spaces clearly.
8. Restart option: Allows users to start a new puzzle or try again.
9. Result display: Shows the final result after completing the game.

### **Tools selection**

For developing the 4×4 Sudoku game project, Java programming language is selected as the main development tool. Java is used to implement the game logic, user input handling and solution validation.

The following tools are selected for this project:

Programming Language: Java

IDE (Integrated Development Environment): NetBeans

GUI Framework: Java Swing (for designing the user interface)

JDK (Java Development Kit): Used for compiling and running the Java application

These tools help in designing, developing, testing and executing the Sudoku Game application effectively.